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Módulo 7. Provedores de Datos Deportivos.

Eventing Opta

Máster en Big Data Deportivo

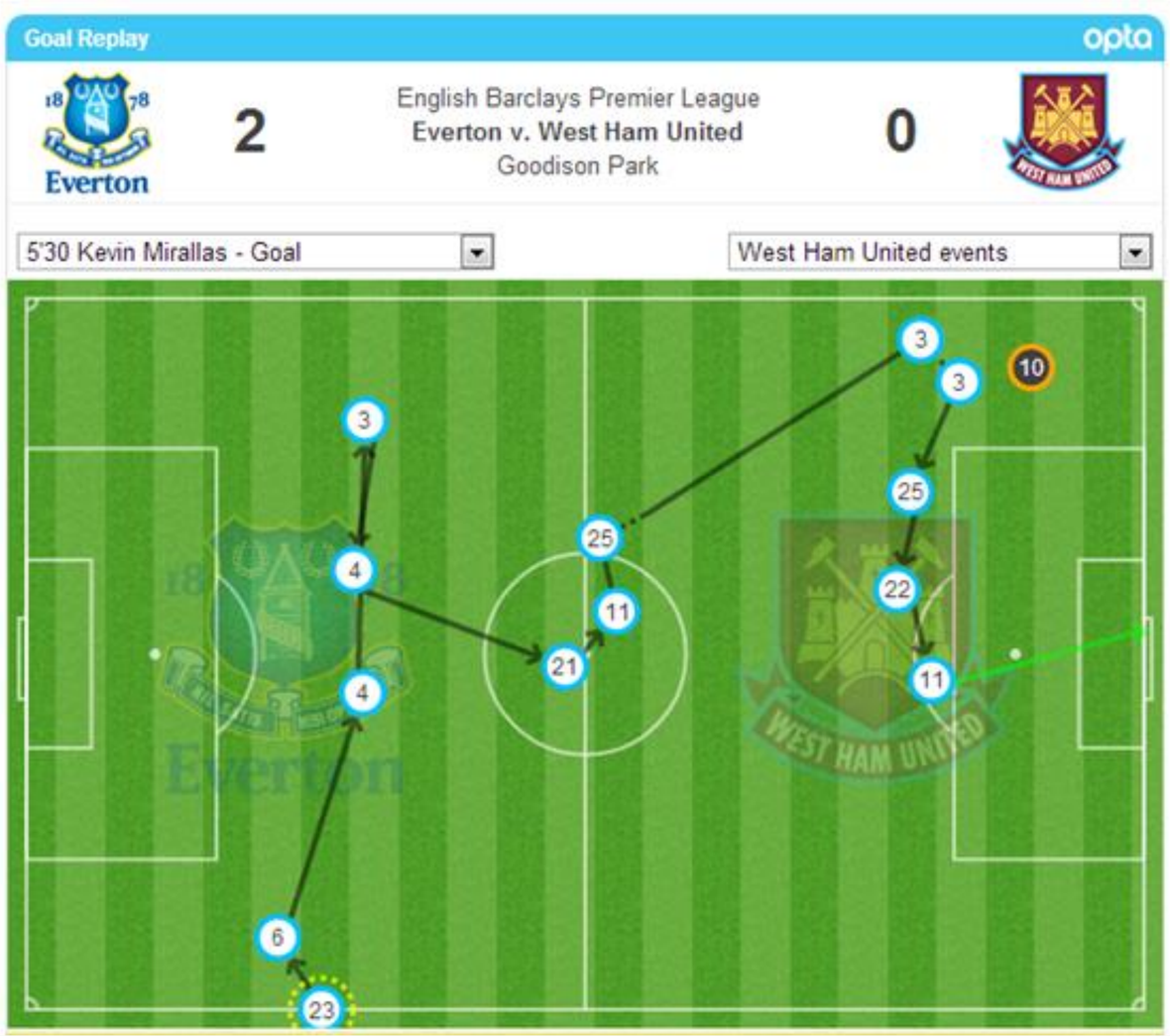
SPORTS DATA CAMPUS

El future del Deporte empieza aquí

Eventing

- Todos los eventos con balón que existen en un Partido incluyendo coordenadas x,y para ubicarlos en el campo
- Entre 1500-3000 eventos por Partido (pases, remates, centros, regates, entradas, paradas, despejes, tarjetas , ...)
- Disponibles para un número muy amplio de Ligas y competiciones.
- Los portales web más conocidos utilizan eventing (FBREF, Understat, Whoscored, Fotmob, Sofascore)
- Proveedores: STATS Perform-Opta, Hudl Wyscout, Statsbomb, BePro, ...

Usos del Eventing - Widgets



05'05	23	Seamus Coleman	Everton	Throw-in
05'06	6	Phil Jagielka	Everton	Pass
05'08	4	Darron Gibson	Everton	Pass
05'12	3	Leighton Baines	Everton	Pass
05'15	4	Darron Gibson	Everton	Pass
05'17	21	Leon Osman	Everton	Pass
05'18	11	Kevin Mirallas	Everton	Pass
05'20	25	Marouane Fellaini	Everton	Pass
05'24	3	Leighton Baines	Everton	Take On
05'24	10	Jack Collison	West Ham United	Unsuccessful Challenge
05'25	3	Leighton Baines	Everton	Pass
05'26	25	Marouane Fellaini	Everton	Pass
05'28	22	Steven Pienaar	Everton	Pass
05'30	11	Kevin Mirallas	Everton	Goal

——— Ball Movement - - - - - - Player Movement

Tracking

- Posición de todos los jugadores y el balón 25 veces por segundo en coordenadas x,y para jugadores x,y,z para balón
- Entre 3 y 4 millones de filas de datos por partido
- Requiere infraestructura big data para su almacenamiento y procesamiento
- Disponibles para un conjunto reducido de competiciones en sus portales oficiales. Alternativa Tracking Televisivo generado por Skill Corner.
- Proveedores: Second Spectrum, TRACAB Chironhego, STATS Perform SportVU , Skill Corner…

Tracking - Ejemplo de un frame da datos

frame	ball_possession	ball_status	team	system_id	jersey_number	x	y	z	speed	object
1893129	A	Dead	10	77		278	-3407	10	157,79	ball
1893129	A	Dead	1	2	10	273	-84	0	2,46	player
1893129	A	Dead	1	29	1	-4310	-315	0	0,97	player
1893129	A	Dead	1	11	26	-1355	-2039	0	0,66	player
1893129	A	Dead	1	28	17	-1170	-864	0	3,30	player
1893129	A	Dead	1	26	8	-897	-2003	0	3,35	player
1893129	A	Dead	1	8	19	-1424	1175	0	1,57	player
1893129	A	Dead	1	10	2	-1696	-2937	0	2,22	player
1893129	A	Dead	1	17	15	-2145	-445	0	2,25	player
1893129	A	Dead	1	16	24	278	-3407	0	1,91	player
1893129	A	Dead	1	6	11	333	-3349	0	5,72	player
1893129	A	Dead	1	1	22	-2189	-1773	0	2,23	player
1893129	A	Dead	0	18	22	-217	-3091	0	3,75	player
1893129	A	Dead	0	25	16	-1034	-1278	0	2,68	player
1893129	A	Dead	0	7	4	471	-3238	0	3,45	player
1893129	A	Dead	0	12	14	657	-842	0	1,95	player
1893129	A	Dead	0	13	6	111	-3267	0	3,01	player
1893129	A	Dead	0	22	10	-823	-2593	0	2,31	player
1893129	A	Dead	0	5	25	3717	-831	0	1,47	player
1893129	A	Dead	0	20	11	-809	-2326	0	1,41	player
1893129	A	Dead	0	21	2	-1725	-1861	0	1,50	player
1893129	A	Dead	0	4	12	397	160	0	4,26	player
1893129	A	Dead	0	3	24	-42	-2480	0	3,43	player

- Frame Frame number
- Ball Possession (Home or Away)
- Ball Status (Alive or Dead)
- Team (0,1 Home and Away, 10 Ball)
- System_id (internal player id)
- Jersey Number
- x,y,z coordinates of the pitch
- speed in m/s or Km/h según proveedor

Eventing – STATS Perform Opta Plantillas Squads F40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       official_club_name="Club Atlético de Madrid SAD" postal_code="28005" region_id="17"
       region_name="Europe" short_club_name="Atlético de Madrid"
       street="Paseo Virgen del Puerto, 67" uID="t175" web_address="www.clubatleticodemadrid.com/">
        <Founded>1903</Founded>
        <Name>Atlético de Madrid</Name>

    <Player uID="p213431">
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            <Position>Defender</Position>
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     </Player>
 </Team>
   </SoccerDocument>
</SoccerFeed>
```


1 Fichero de Plantillas por cada competición y temporada
Ejemplo La Liga 2020/2021

Team

- uID→ Para cruzar por ejemplo con F24 team_id
- official_club_name
- short_club_name

Player

- uID→ Para cruzar con F24 player_id
- Name

Eventing STATS Perform – Tipos de Eventos

Appendix 1 - Event types

Event ID	Name	Description
1	Pass	Any pass attempted from one player to another - free kicks, corners, throw-ins, goal kicks and goal assists
2	Offside Pass	Attempted pass made to a player who is in an offside position
3	Take On	Attempted dribble past an opponent (excluding when qualifier 211 is present as this is 'overrun' and is not always a duel event)
4	Foul	This event ID shown when a foul is committed resulting in a free kick
5	Out	Shown each time the ball goes out of play for a throw-in or goal kick
6	Corner Awarded	Ball goes out of play for a corner kick
7	Tackle	Tackle = dispossesses an opponent of the ball - Outcome 1 = win & retain possession or out of play, 0 = win tackle but not possession
8	Interception	When a player intercepts any pass event between opposition players and prevents the ball reaching its target. Cannot be a clearance.
9	Turnover	Unforced error / loss of possession - i.e. bad control of ball – Replaced with Unsuccessful Touch + Overrun in recent seasons
10	Save	Goalkeeper event; saving a shot on goal. Can also be an outfield player event with qualifier 94 for blocked shot.
11	Claim	Goalkeeper event; catching a crossed ball
12	Clearance	Player under pressure hits ball clear of the defensive zone or/and out of play
13	Miss	Any shot on goal which goes wide or over the goal
14	Post	Whenever the ball hits the frame of the goal
15	Attempt Saved	Shot saved - this event is for the player who made the shot. Qualifier 82 can be added for blocked shot.
16	Goal	All goals
17	Card	Bookings - will have red, yellow or 2nd yellow qualifier plus a reason
18	Player Off	Player is substituted off
19	Player on	Player comes on as a substitute
20	Player retired	Player is forced to leave the pitch due to injury and the team have no substitutions left
21	Player returns	Player comes back on the pitch
22	Player becomes goalkeeper	When an outfield player has to replace the goalkeeper
23	Goalkeeper becomes player	Goalkeeper becomes an outfield player
24	Condition change	Change in playing conditions
25	Official change	Referee or linesman is replaced
27	Start delay	Used when there is a stoppage in play such as a player injury
28	End delay	Used when the stoppage ends and play resumes
30	End	End of a match period
32	Start	Start of a match period
34	Team set up	Team line up - qualifiers 30, 44, 59, 130, 131 will show player line up and formation
35	Player changed position	Player moved to a different position but the team formation remained the same
36	Player changed Jersey number	Player is forced to change jersey number, qualifier will show the new number
37	Collection End	Event 30 signals end of half. This signals end of the match and thus data collection.
38	Temp_Goal	Goal has occurred but it is pending additional detail qualifiers from Opta. Will change to event 16.
39	Temp_Attempt	Shot on goal has occurred but is pending additional detail qualifiers from Opta. Will change to event 15.
40	Formation change	Team alters its formation
41	Punch	Goalkeeper event; ball is punched clear
42	Good skill	A player shows a good piece of skill on the ball – such as a step over or turn on the ball
43	Deleted event	Event has been deleted – the event will remain as it was originally with the same ID but will be resent with the type altered to 43.
44	Aerial	Aerial duel – 50/50 when the ball is in the air – outcome will represent whether the duel was won or lost
45	Challenge	When a player fails to win the ball as an opponent successfully dribbles past them
47	Rescinded card	This can occur post match if the referee rescinds a card he has awarded
49	Ball recovery	When a player takes possession of a loose ball
50	Dispossessed	Player is successfully tackled and loses possession of the ball
51	Error	Mistake by player losing the ball. Leads to a shot or goals as described with qualifier 169 or 170
52	Keeper pick-up	Goalkeeper event; picks up the ball
53	Cross not claimed	Goalkeeper event; cross not successfully caught
54	Smother	Goalkeeper event; comes out and covers the ball in the box winning possession
55	Offside provoked	Awarded to last defender when an offside decision is given against an attacker
56	Shield ball opp	Defender uses his body to shield the ball from an opponent as it rolls out of play
57	Foul throw-in	A throw-in not taken correctly resulting in the throw being awarded to the opposing team
58	Penalty faced	Goalkeeper event; penalty by opposition
59	Keeper Sweeper	When keeper comes off his line and/or out of his box to clear the ball
60	Chance missed	Used when a player does not actually make a shot on goal but was in a good position to score and only just missed receiving a pass
61	Ball touch	Used when a player makes a bad touch on the ball and loses possession. Outcome 1 – ball simply hit the player unintentionally. Outcome 0 – Player unsuccessfully controlled the ball.
63	Temp_Save	An event indicating a save has occurred but without full details. Event 10 will follow shortly afterwards with full details.
64	Resume	Match resumes on a new date after being abandoned mid game
65	Contentious referee decision	Any major talking point or error made by the referee – decision will be assigned to the relevant team
66	Possession Data	Possession event will appear every 5 mins **No longer recorded in the feed**
67	50/50	2 players running for a loose ball - GERMAN ONLY. Outcome 1 or 0. | No longer collected as of June 2017
68	Referee Drop Ball	Delay - ref stops - this to event given to both teams on restart. No Outcome
69	Failed to Block	Attempt to block a shot or pass - challenge lost. Put Through (qualifiers 266) is the winning duel event. | Collected for DFL competitions only between 2013/14 and 2016/17
70	Injury Time Announcement	Injury Time awarded by Referee
71	Coach Setup	Coach Type; 1,2,18,30,32,54,57,58,59
72	Caught Offside	New event to just show player who is offside instead of offside pass event
73	Other Ball Contact	This is an automated extra event for DFL. It comes with a tackle or an interception and indicates if the player who made the tackle/interception retained the ball after this action or if the tackle/interception was a single ball touch (other ball contact with type "interception", type "Defensive Clearance" or type " TackleRetainedBall").
74	Blocked Pass	Similar to interception but player already very close to ball
75	Delayed Start	Match start delayed
76	Early end	The match has had an early end
77	Player Off Pitch	Event indicating that a player is now off the pitch
79	Coverage interuption	Used to identify when our analysis of a game has been interupted. outcome 0 = start of interruption, outcome 1 = end of interruption

Eventing STATS Perform – Algunos Qualifiers

Appendix 2 - Qualifier types

ID	Pass Events	Values	Description (ASSOCIATED TO EVENT TYPE 1)
1	Long ball		Long pass over 32 metres
2	Cross		A ball played in from wide areas into the box
3	Head pass		Pass made with a players head
4	Through ball		Ball played through for player making an attacking run to create a chance on goal
5	Free kick taken		Any free kick; direct or indirect
6	Corner taken		All corners. Short corners will feature qualifier 6 but be missing qualifier 2 (cross).
7	Players caught offside	Players caught offside	Player who was in an offside position when pass was made.
8	Goal disallowed		Pass led to a goal disallowed for a foul or offside
106	Attacking Pass		A pass in the opposition's half of the pitch | No longer collected as of June 2016
107	Throw-in		Throw-in taken
140	Pass End X	0_100	The x pitch coordinate for the end point of a pass - See Appendix 11
141	Pass End Y	0_100	The y pitch coordinate for the end point of a pass - See Appendix 11
152	Direct		A direct free kick.
155	Chipped		Pass which was chipped into the air
156	Lay-off		Pass where player laid the ball into the path of a teammates run
157	Launch		Pass played from a player's own half up towards front players. Aimed to hit a zone rather than a specific player
168	Flick-on		Pass where a player has "flicked" the ball forward using their head
193	Goal Measure		Essentially a Goal Rating from 1-5 but is only used for a limited about of comps. This was collected from 2013-2015 but is no longer used
195	Pull Back		Player in opposition's penalty box reaches the by-line and passes (cuts) the ball backwards to a teammate
196	Switch of play		Any pass which crosses the centre zone of the pitch and in length is greater than 60 on the y axis of the pitch
210	Assist		The pass was an assist for a shot. The type of shot then dictates whether it was a goal assist or just key pass.
212	Length	Dynamic - length of pass in metres	The estimated length the ball has travelled during the associated event.
213	Angle	0 to 6.28 (Radians)	The angle the ball travels at during an event relative to the direction of play. Shown in radians.
218	2nd assist		Pass was deemed a 2nd assist - created the opportunity for another player to assist a goal
219	Players on both posts		Assigned to a pass event with qualifier 6 indicating if there were defensive players on both posts when a corner was taken | No longer collected as of June 2016
220	Player on near post		Assigned to a pass event with qualifier 6 indicating if there was a defensive player on only the near post when a corner was taken | No longer collected as of June 2016
221	Player on far post		Assigned to a pass event with qualifier 6 indicating if there was a defensive player on only the far post when corner was taken | No longer collected as of June 2016
222	No players on posts		Assigned to a pass event with qualifier 6 indicating if there were no defensive players on either post when a corner was taken | No longer collected as of June 2016
223	In-swinger		Corner was crossed into the box swerving towards the goal
224	Out-swinger		Corner was crossed into the box swerving away from the goal
225	Straight		Corner was crossed into the box with a straight ball flight
236	Blocked Pass		Similar to interception but player already very close to ball - instead of touch event in past. Or assigned to the pass to indicate it was blocked
238	Fair Play		Ball kicked out of play for injury etc
240	GK Start		Automated qualifier which appears on all GK passes after keeper collects ball with his hands and then puts it on the ground.
241	Indirect		Shows if indirect freekick is taken
266	Put Through		Attacking player has passed/shot the ball past a defending player who attempted to stop the ball. A "put through " is always linked with a "failed to block" event. | Collected for DFL competitions only between 2013/14 and 2016/17
279	Kick Off	S = kick-off to start a period (first half, second half, first half extra-time, second half extra-time) G = kick-off following a goal	Starting pass - to enable exclusion from passing %
278	Tap		Ball tapped to other player (i.e. for free-kick shot)
287	Over-arm		Indicates over-arm throw out by the goalkeeper
345	Overhit Cross		A cross which clearly went over the intended target area
346	Next Event - Goal Kick		Used with Event 5. Indicates the next event will be a goal-kick
347	Next Event - Throw-In		Used with Event 5. Indicates the next event will be a throw-in

...

Eventing – STATS Perform Opta Eventos Feed F24

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  </Game>
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Game (información del partido)

Event

- type_id → Ver tabla tipos de eventos
- period_id → 1ª Parte, 2ª Parte
- min → Minuto
- sec → Segundo
- player_id → Id del jugador ir a plantillas para ver su nombre
- team_id → id del equipo
- outcome → 0 o 1 en algunos eventos indica 1 completado, 0 no completado
- x → ver dimensiones campo
- y → ver dimensiones campo

Qualifier (Q)

- qualifier_id → Ver tabla con qualifiers
- value → valor asociado al qualifier

Eventing – STATS Perform – Evento Pase

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<Event id="2253496891" event_id="6" type_id="1" period_id="1" min="0" sec="4" player_id="213431"
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        <Q id="2693873611" qualifier_id="141" value="70.3" />

</Event>
```

Event

- type_id→ 1 - Pase
- period_id→ 1 - 1ª Parte
- min → Minuto 0
- sec → Segundo 4
- player_id → id jugador (Mario Hermoso)
- team_id→ id del equipo (Atlético Madrid)
- outcome→ 0 - pase no completado
- X,y→ ver dimensiones campo

Qualifiers

- qualifier_id→ 1 Long ball - Long pass over 32 metres
- qualifier_id→ 140 Pass End X
- qualifier_id→ 141 Pass End Y
- qualifier_id→ 155 Chipped Pass which was chipped into the air
- qualifier_id→ 212 32 metros Length The estimated length in metres the ball has travelled during the associated event.
- qualifier_id→ 213 Angle (0 to 6.28 Radians) The angle the ball travels at during an event relative to the direction of play.
- qualifier_id→ 56 Zone (Back, Left, Center, Right)

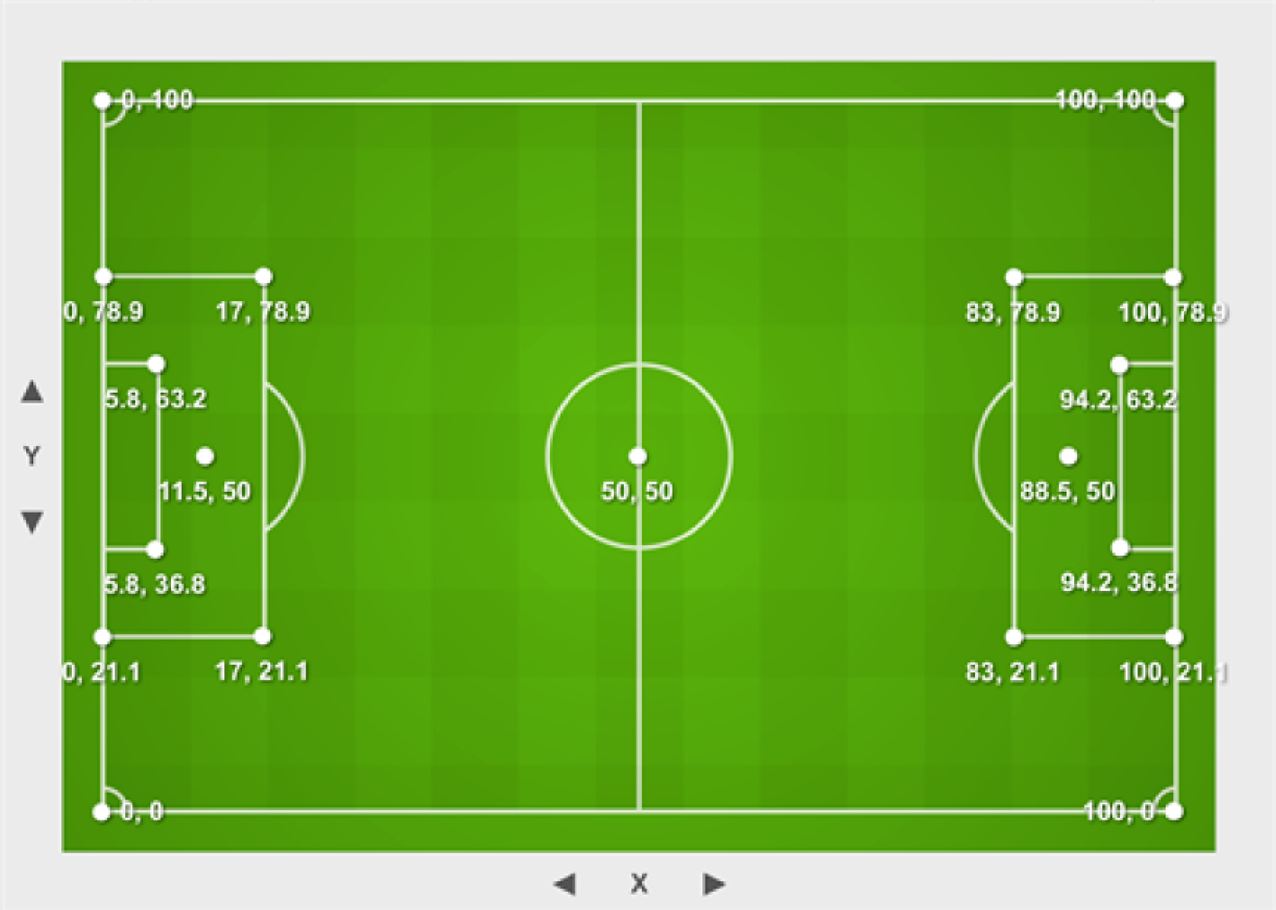
Eventing – STATS Perform – Evento Pase

Inicio Pase

Final Pase

Eventing – STATS Perform Dimensiones campo

This is based on the direction of attack always being from left to right on the x-axis, regardless of the team or period of the game, the defending goal line is always where x=0 and the attacking where x=100. All measurements for events given in the feed should therefore be taken in relation to percentages.



Eventing – STATS Perform Dimensiones Portería

Goalmouth y,z(Qualifier_id 102,103) and Zones of the Goalmouth 76-87

EVENT - Goal, Attempt Saved		
Qualifier	**end_y**	**end_z**
Low Left	(51.8 <= end_y <= 54.8)	(0 <= end_z <= 20)
High Left	(51.8 <= end_y <= 54.8)	(20 <= end_z <= 38)
Low Centre	(48.2 <= end_y <= 51.8)	(0 <= end_z <= 20)
High Centre	(48.2 <= end_y <= 51.8)	(20 <= end_z <= 38)
Low Right	(45.2 <= end_y <= 48.2)	(0 <= end_z <= 20)
High Right	(45.2 <= end_y <= 48.2)	(20 <= end_z <= 38)
EVENT - Post		
Qualifier	**end_y**	**end_z**
Left	(54.8 <= end_y <= 55.8)	(0 <= end_z <= 38)
High	(44.2 <= end_y <= 55.8)	(38 <= end_z <= 42)
Right	(44.2 <= end_y <= 45.2)	(0 <= end_z <= 38)
EVENT - Miss		
Qualifier	**end_y**	**end_z**
Close left	(55.8 <= end_y <= 59.3)	(0 <= end_z <= 40)
Close High Left	(55.8 <= end_y <= 59.3)	(40 <= end_z <= 60)
Close Right	(40.7 <= end_y <= 44.2)	(0 <= end_z <= 40)
Close High Right	(40.7 <= end_y <= 44.2)	(40 <= end_z <= 60)
Close High	(44.2 <= end_y <= 55.8)	(42 <= end_z <= 60)
Left	(59.3 <= end_y <= 100)	(0 <= end_z <= 40)
Right	(0 <= end_y <= 40.7)	(0 <= end_z <= 40)
HighLeft	(55.8 <= end_y <= 100)	(60 <= end_z <= 100)
HighRight	(0 <= end_y <= 44.2)	(60 <= end_z <= 100)
High	(44.2 <= end_y <= 55.8)	(60 <= end_z <= 100)

Eventing – STATS Perform Periodos

Appendix 10 - Period ID list

Period	Period_id
First half	1
Second Half	2
First period of extra time	3
Second period of extra time	4
Penalty shoot out	5
Half-time	10
Post-game	14
Pre-game	15
Pre-match	16

Eventing – STATS Perform – Zonas del Campo

When Qualifier 56 is shown with an event it will also display a value which can be right, left, centre or back. The diagram below highlights what these zones are on the pitch.

Shot Location Descriptors

Qualifiers 16, 17, 18, 19 and 60 - 71 refer to areas of the pitch. The below diagram illustrates
where these zones are.

Eventing – STATS Perform – Descriptores de Remates

Parte del cuerpo

Body Part

15	Head		Any event where the player used their head such as a shot or a clearance
72	Left footed		Player shot with their left foot
20	Right footed		Player shot with right footed
21	Other body part		Shot was neither via a player's head or foot for example knee or chest

Patrón o Fase de Juego

Pattern of Play

22	Regular play		Shot during open play as opposed to from a set play
23	Fast break		Shot occurred following a fast break situation
24	Set piece		Shot occurred from a crossed free kick
25	From corner		Shot occurred from a corner
26	Free kick		Shot occurred directly from a free kick
29	Assisted		Indicates that there was a pass (assist) from another player to set up the goal opportunity
55	Related event ID	Event_id	This will appear for goals or shots, the related event_id will be that of the assist and thus show the assisting player ID
96	Corner situation		Pass or shot event in corner situation. 25 is used when the goal is direct from corner, 96 relates to 2nd phase attack - NO LONGER USED
97	Direct free		26 will be used for shot directly from a free kick. 97 only used with Opta GoalData (game system 4) but not with full data.
112	Scramble		Goal where there was a scramble for possession of the ball and the defence had an opportunity to clear | No longer collected as of June 2016
154	Intentional assist		Shot from an intentional assist i.e. The assisting player intended the pass, no deflection etc
160	Throw-in set piece		Shot came from a throw-in set piece
216	2nd related event ID	Event_id	If there was a 2nd assist, i.e a pass to create the opportunity for the player making the assist. MLS and German Bundesliga 1 & 2.
233	Opposite related event ID		Used for any event where there is the same event for both teams with outcome 1 or 0. This relates the 2 event together, for example aerial duels

Eventing – STATS Perform – Evento Gol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team_id="175" outcome="1" x="87.0" y="37.3" timestamp="2021-01-12T20:49:53.676"
last_modified="2021-01-13T01:23:10" version="1610500990729">
      <Q id="2693911299" qualifier_id="56" value="Center" />
      <Q id="2693914393" qualifier_id="17" />
      <Q id="2693913825" qualifier_id="230" value="98.4" />
      <Q id="2693913817" qualifier_id="55" value="169" />
      <Q id="2693913811" qualifier_id="80" />
      <Q id="2693912395" qualifier_id="374" value="2021-01-12 20:49:52.726" />
      <Q id="2694133803" qualifier_id="395" value="1.2" />
      <Q id="2693913813" qualifier_id="29" />
      <Q id="2694204901" qualifier_id="178" />
      <Q id="2693912397" qualifier_id="375" value="16:07.163" />
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      <Q id="2693913821" qualifier_id="102" value="45.8" />
      <Q id="2693913827" qualifier_id="231" value="48.0" />
      <Q id="2693913815" qualifier_id="154" />
      <Q id="2693911295" qualifier_id="22" />
      <Q id="2693912391" qualifier_id="72" />
      <Q id="2693913823" qualifier_id="103" value="2.5" />
```


Event

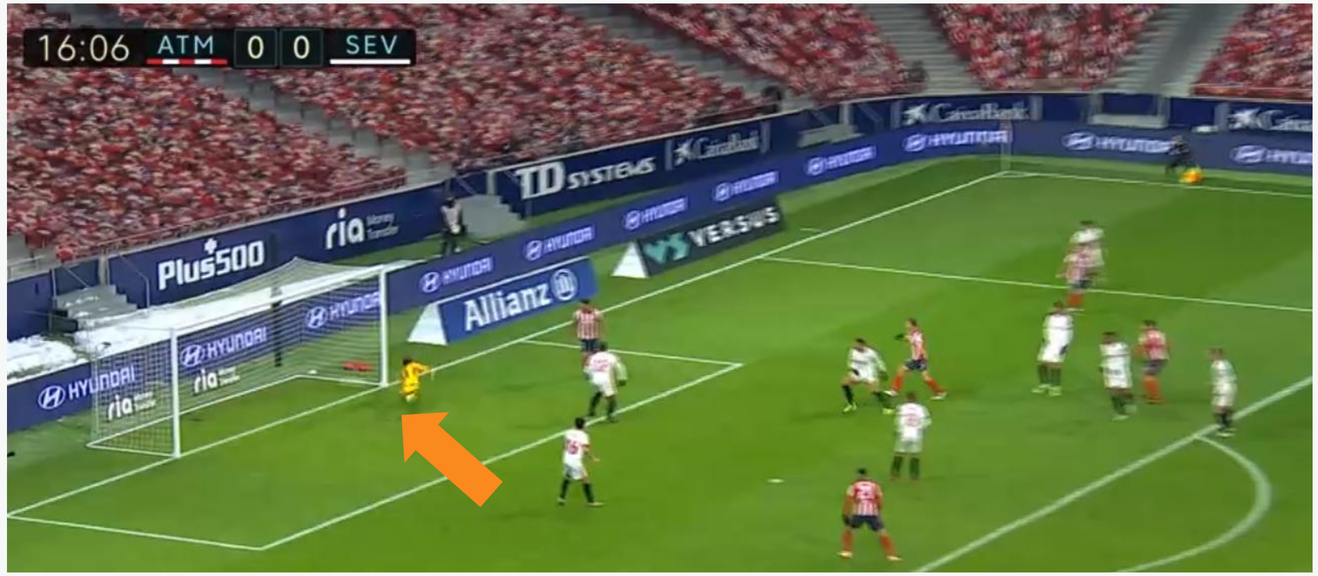
- type_id → 16 - Gol
- period_id → 1 - 1ª Parte
- min → Minuto 16
- sec → Segundo 8
- player_id → id jugador (Ángel Correa)
- team_id → id del equipo (Atlético Madrid)
- outcome → 1
- x,y → ver dimensiones campo

Qualifiers

- qualifier_id → 17 Box Centre- Shot Location Descriptor
- qualifier_id → 102 Goalmouth y Coordinate of where a shot crossed goal line
- qualifier_id → 103 Goalmouth z Coordinate of where a shot crossed goal line
- qualifier_id → 80 Goalmouth zone - Low Right
- qualifier_id → 230 GK X position when goal or shot hit post
- qualifier_id → 231 GK Y position when goal or shot hit post
- qualifier_id → 29 Assisted (shot with a pass assist) from another player to set up goal opportunity
- qualifier_id → 154 Intentional Assist Shot from an intentional assist
- qualifier_id → 56 Zone (Back, Left, Center, Right)
- qualifier_id → 55 Related Event Id
- qualifier_id → 22 Regular Play
- qualifier_id → 72 Left foot Body part

SPORTS DATA CAMPUS

Eventing – STATS Perform – Evento Gol



16:05 ATM 0 0 SEV
TD SYSTEMS
CaixaBank
HYUNDAI
Plus500
ria Money Transfer
Allianz
VERSUS

16:06 ATM 0 0 SEV
TD SYSTEMS
HYUNDAI
Plus500
ria Money Transfer
Allianz
VERSUS

Contenido del Notebook

Flujo de principales dataframes que se utilizarán en la sesión

1 fichero por partido ejemplo **Atlético de Madrid - Sevilla Temporada 2020-21**
Dataframe con todos los **pases del partido** fichero F24.xml

df_pases

1 fichero por Temporada y Competición
Dataframe con todos los **nombres de los jugadores** fichero F40 Squads.xml

df_nombre_jugadores

df_pases_final

1 Dataframe con todos los **pases del partido** y **nombres de jugadores**

Filtro Jugador

Filtro Equipo y Filtro Pases Largos

df_pases_jugador

Filtro todos los pases de un jugador

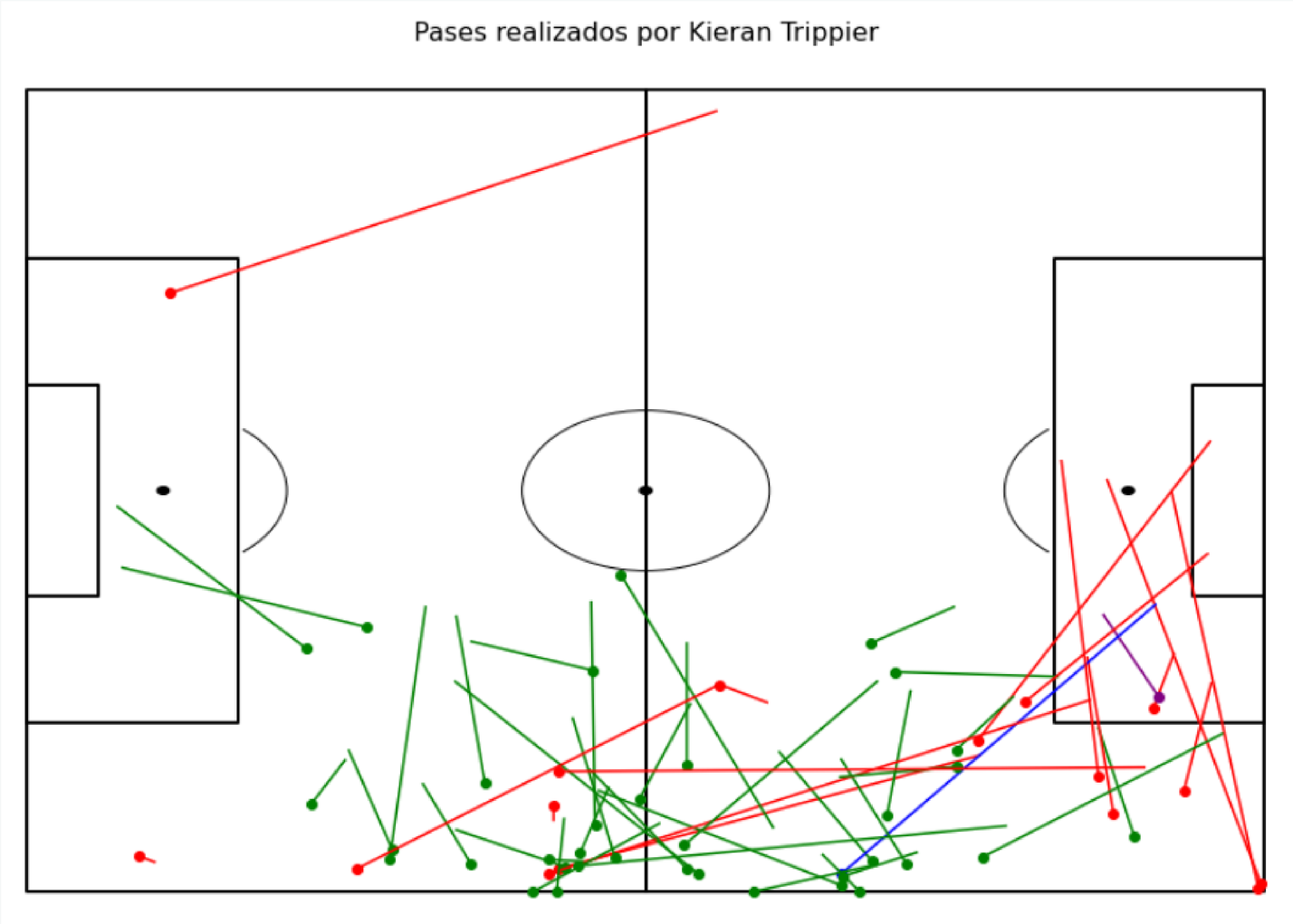
df_pases_equipo

Filtro pases largos (>40m) de un equipo

Contenido del Notebook

Todos los Pases de Kieran Trippier

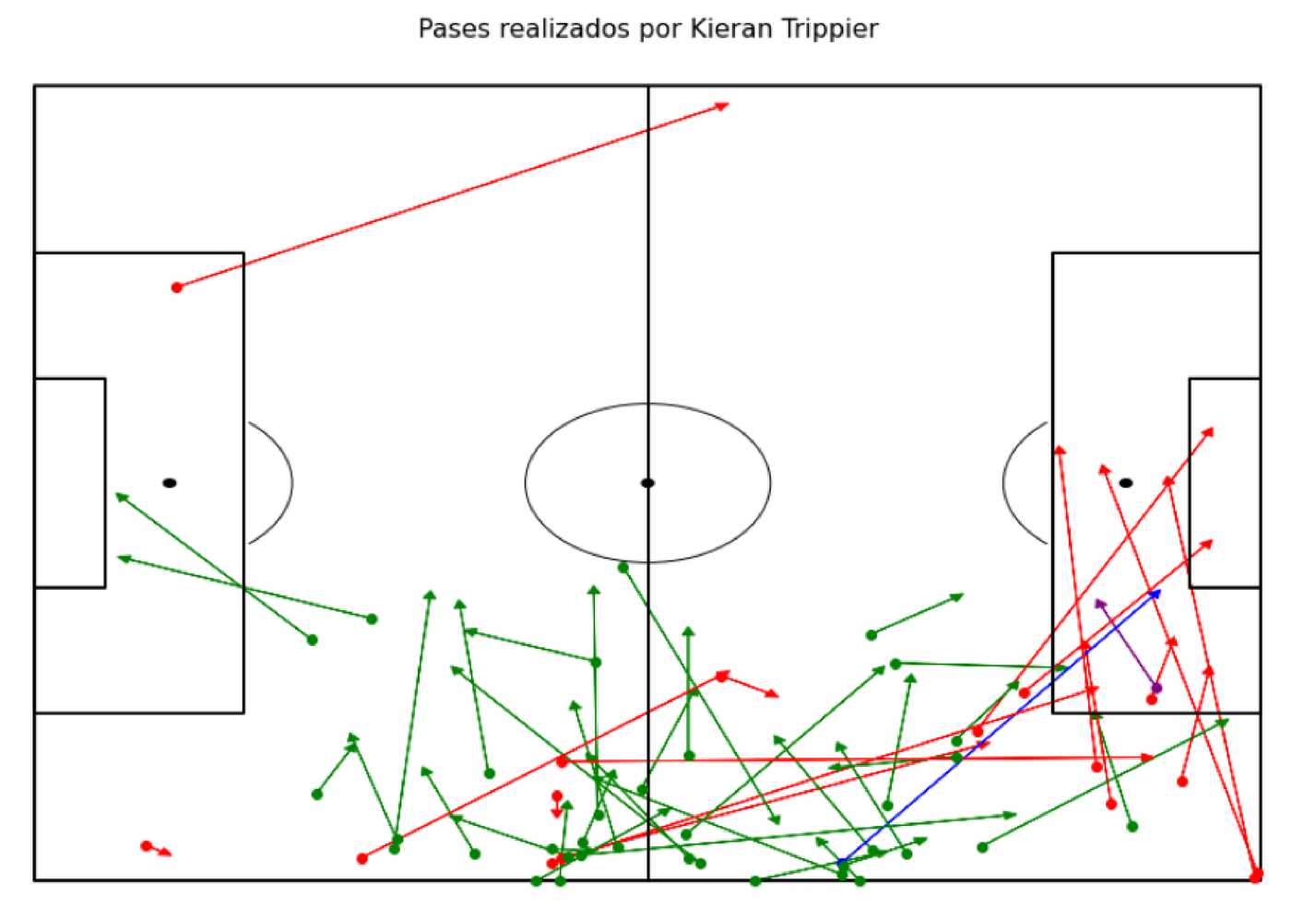
- Asistencia Morado
- Azul Pase Clave
- Verde Pase Completado
- Rojo Pase No Completado



Contenido del Notebook

Todos los Pases de Kieran Trippier con flechas

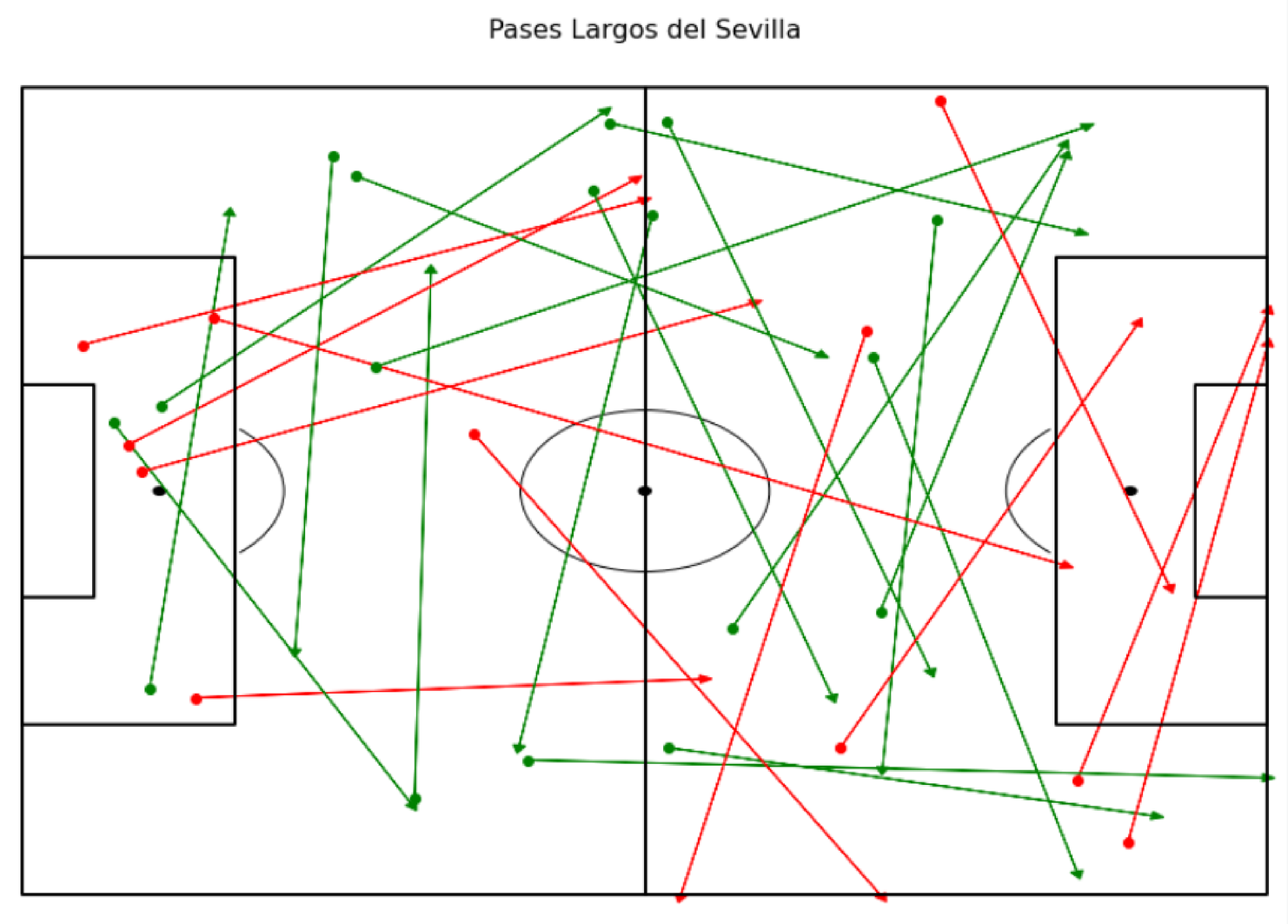
- Asistencia Morado
- Azul Pase Clave
- Verde Pase Completado
- Rojo Pase No Completado



Contenido del Notebook

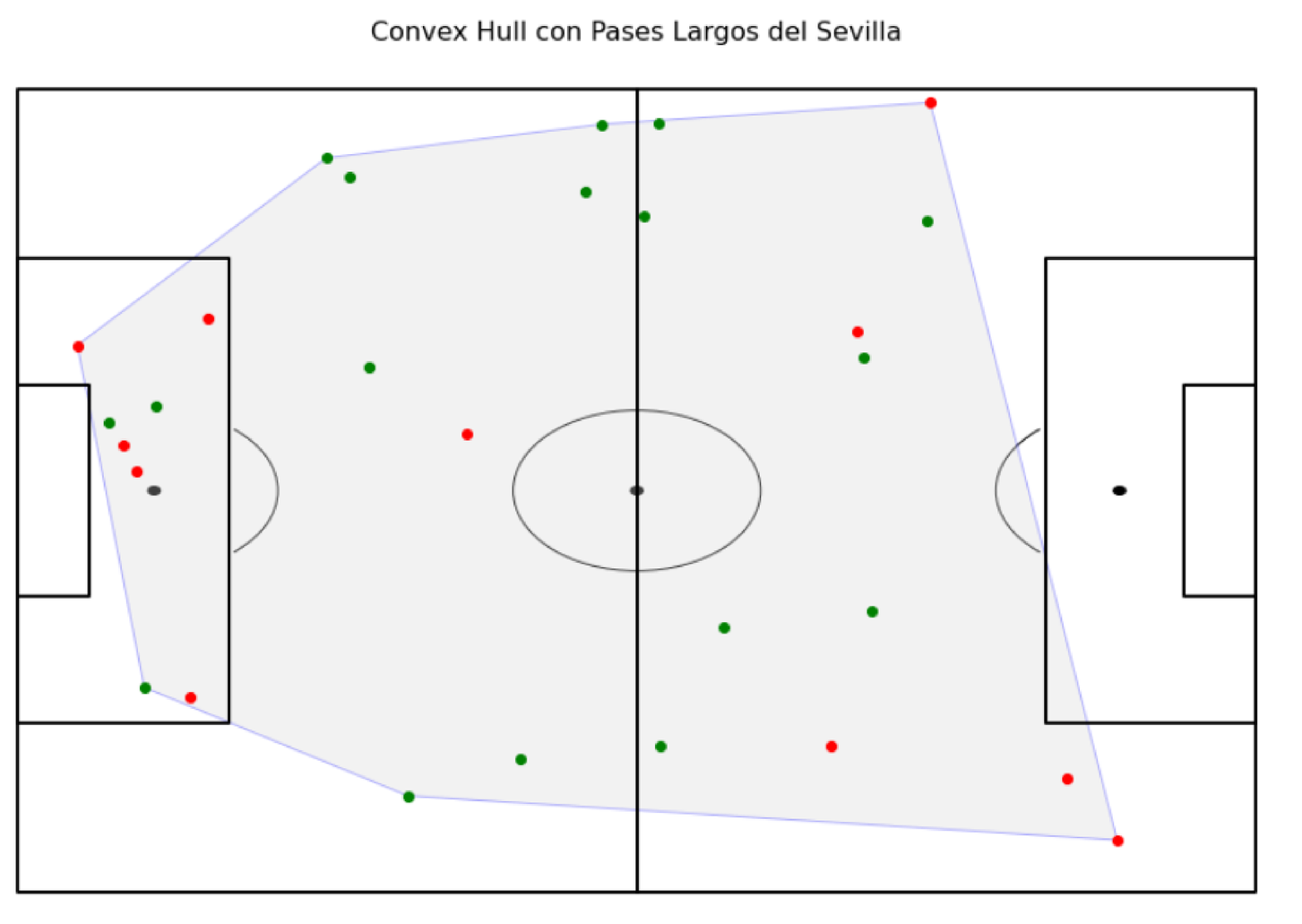
Todos los Pases Largos del Sevilla FC

- Asistencia Morado
- Azul Pase Clave
- Verde Pase Completado
- Rojo Pase No Completado



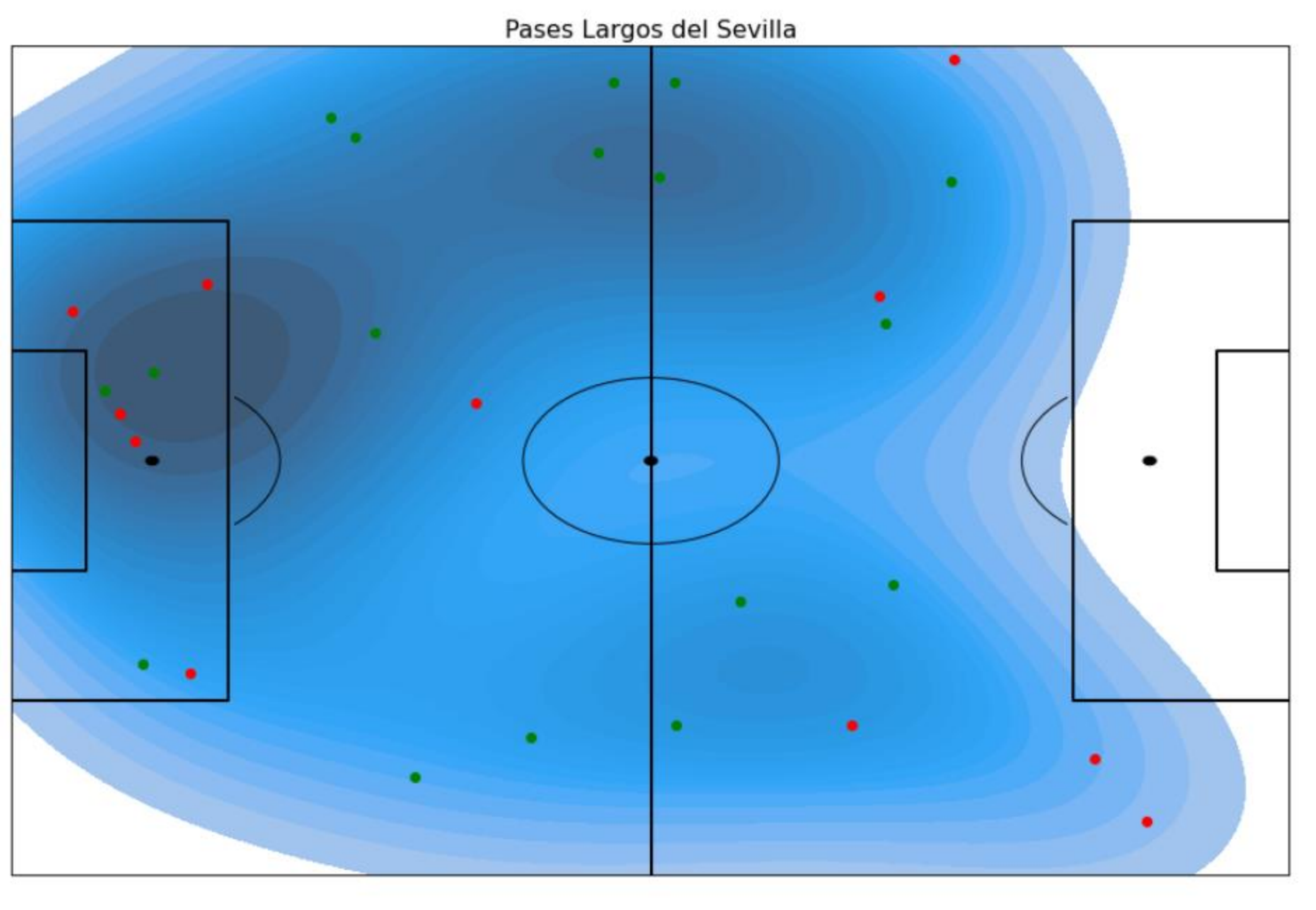
Contenido del Notebook

Convex Hull con todos los Pases Largos del Sevilla FC



Contenido del Notebook

Mapa de Calor **KDEPlot** con todos los Pases Largos del Sevilla FC



GRACIAS POR ASISTIR

SPORTS DATA
CAMPUS

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